TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 4

5-MINUTE REVISION

Central Tendency I

From the chapter's own revision points, formulae and questions.

BOOK-ONLY EDITION · Verified against the textbook

Selected from Chapter 4 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- Mean → arithmetic average of observations.
- Median → middle value after arranging observations.
- Mode → most frequently occurring value.
- Mean → uses all observations.
- Median → based on position.

Quick revision · 2

- Mode → based on frequency.
- Mean is affected by extreme values.
- Median is less affected by extreme values.
- Mode is useful for identifying the most common category or value.
- Mean + constant → mean increases by the same constant.

Quick revision · 3

- Mean × constant → mean is multiplied by the same constant.
- Adding a constant to every observation does not change the standard deviation.
- Multiplying every observation by a constant multiplies the standard deviation by the absolute value of that constant.
- Sum of deviations from the arithmetic mean = 0.
- Standard deviation = 0 → all observations are identical.

Quick revision · 4

- For positive observations, $AM \geq GM \geq HM$.
- Symmetric unimodal distribution with a finite mean and a central mode → mean = median = mode.
- In many familiar positively skewed unimodal distributions, mean > median > mode; this is not a universal theorem.
- In many familiar negatively skewed unimodal distributions, mean < median < mode; this is not a universal theorem.

Important formulae · 1

Arithmetic mean:

$$\bar{x} = \frac{\sum x_i}{n}$$

Arithmetic mean for a frequency distribution:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

Assumed mean method:

$$\bar{x} = A + \frac{\sum f_i d_i}{\sum f_i}$$

where

$$d_i = x_i - A$$

Important formulae · 2

Coding method:

$$\bar{x} = A + C\frac{\sum f_i u_i}{\sum f_i}$$

where

$$u_i = \frac{x_i - A}{C}$$

Weighted mean:

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

Median for odd n:

$$Md = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ observation}$$

Important formulae · 3

Median for even n:

$$Md = \frac{1}{2}\left[\left(\frac{n}{2}\right)^{\text{th}} \text{ observation} + \left(\frac{n}{2}+1\right)^{\text{th}} \text{ observation}\right]$$

Median for grouped data:

$$Md = L + \left(\frac{\frac{N}{2} - F}{f}\right)C$$

Mode for grouped data:

$$Mo = L + \left(\frac{f_m - f_p}{2f_m - f_p - f_s}\right)C$$

Important formulae · 4

Relationship among arithmetic mean, geometric mean, and harmonic mean:

$$AM \geq GM \geq HM$$

Worked example from the chapter

Find the mean of the numbers 2, 4, 7, 8, 11, 12

Solution

$$\overline{x} = \frac{2+4+7+8+11+12}{6} = \frac{44}{6} = 7.33$$

Self-check from the chapter

1. The sum of observations divided by their number is called the arithmetic _____. **[Mean]**
2. The sum of deviations from the arithmetic mean is _____. **[Zero]**
3. The average calculated using assigned weights is called the _____ mean. **[Weighted]**
4. The middle value of an ordered dataset is called the _____. **[Median]**

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